Last name _____

First name _____

LARSON—MATH 310–HOMEWORK WORKSHEET 11
Matrix Pseudo Inverse Calculations

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Matrix Inverse

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

1. We know $A = QR$ for an orthogonal matrix Q and an upper-triangular matrix R. Find Q.
2. We know orthogonal matrices Q are invertible with inverse $Q^{-1} = Q^T$. So, $Q^T A = R$. Use this fact to find R (In particular $R_{i,j} = (Q^T A)_{i,j}$. So $R_{i,j}$ is the inner product of $\vec{q}_i$ and $\vec{a}_j$).
3. Check that A has linearly independent columns (for a square matrix that's enough to guarantee that A has an inverse A^{-1}).
4. We know $A^{-1} = (QR)^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$. So to find A^{-1} we just need R^{-1}. Since R is upper-triangular with no zero entry on its diagonal we know it has an inverse. Solve
$$R\left[\vec{x}_1\vec{x}_2\right] = \left[\vec{e}_1\vec{e}_2\right],$$
by solving $R\vec{x}_1 = \vec{e}_1$ and $R\vec{x}_2 = \vec{e}_2$. (These correspond to systems that can be backsolved.)
5. Let $R^{-1} = \left[\vec{x}_1\vec{x}_2\right]$. Check that $RR^{-1} = I$.
6. Now find A^{-1} (we know $A^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$).
7. Check that $AA^{-1} = I$.

(over)

Matrix Pseudo Inverse

Let

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$

8. Check that B has linearly independent columns (then we know that B^TB has an inverse $(B^TB)^{-1}$).

   We showed that if B has linearly independent columns then it has a left inverse (so there is a matrix X where $XB = I$). In fact, we showed that this *pseudo-inverse* of B is $(B^TB)^{-1}B^T$.

9. Find B^TB.

10. Use your previous results to find $(B^TB)^{-1}$.

11. Let $X = (B^TB)^{-1}B^T$. Find X.

12. Check that X is the pseudo-inverse (check that $XB = I$).